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Double trouble: host behaviour influences and is influenced by co-infection with parasites

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Parasitism is increasingly seen as an ecological factor contributing to behavioural variation among individuals. Yet, the causal direction of the relationship between animal personality and parasites remains unclear. We measured behavioural traits (i.e. exploration, activity, boldness) in pumpkinseed sunfish, *Lepomis gibbosus*, before and after an experimental infection using cages in a lake where sunfish were naturally exposed to trematode and cestode infection for 1 month. Despite our initial assumptions (i.e. that all individuals would have the same risk of infection within a cage), we found that initial behavioural traits strongly influenced infection susceptibility: initially bolder and less active fish acquired a higher density of trematode and cestode parasites during the infection period. Following infection, fish body condition decreased with increasing cestode density, suggesting that infection is costly to hosts. Body condition was positively correlated with distance moved, a measure of activity, regardless of individual infection status. The repeatability of exploration and activity behaviour and the strength of the activity–exploration correlation (i.e. behavioural syndrome) were reduced following parasite exposure. Distance moved and trematode density were negatively correlated, suggesting that this infection decreases host activity levels. Since trematodes have a complex life cycle with piscivorous birds as a final host, a decrease in activity following infection may make infected fish more susceptible to bird predation, benefiting the parasite. Our results highlight the close links between behaviour and parasitism. We propose that two mechanisms may simultaneously operate: initial host behaviour influences their risk of infection, and infection introduces variation in the behavioural traits of infected hosts.

It is well established that interindividual behavioural differences consistent over time and contexts (i.e. personality) exist within wild animal populations (Barber & Dingemanse, 2010). Ecological factors, including predation and resource competition, can act as drivers of behavioural variation in a variety of systems (Bell & Sih, 2007; Bucklaew & Dochtermann, 2021; Dingemanse & Réale, 2005). Similarly, state-dependent traits such as body condition can also impact behavioural traits (Moiron et al., 2019) and have been linked to parasite infection (Debeffe et al., 2014; Thelamon, 2023). By contrast, the overall importance of parasites, organisms that live and use resources on or in a host (Goater et al., 2014), in driving behavioural differences in wild populations remains unclear despite growing interest in the subject (Barber et al., 2017; Dubois & Binning, 2022).

Specifically, a growing number of studies have documented correlations between parasite infection and animal behaviour (Barber & Dingemanse, 2010; Kortet et al., 2010), suggesting that infected individuals behave differently than uninfected ones in a range of behaviours (e.g. risk taking, exploration, predator avoidance) (Barber et al., 2017; Berdoy et al., 2000; Poulin, 1994). Two nonmutually exclusive hypotheses can explain these trends. First, infection with parasites may cause changes in host behaviour (Poulin, 1995). Direct modification of behaviour by parasites can occur via parasite manipulation, a phenomenon often seen in hosts infected by parasites with complex life cycles (i.e. requiring several intermediate hosts) (Barber & Scharsack, 2010; Lafferty, 1999; Parker et al., 2009). For example, infection by *Toxoplasma gondii* changes the behaviour of host rats (*Rattus norvegicus*) that become attracted, rather than repelled, by the small of cat urine. This altered behaviour makes it more likely that infected rats are depredated by cats, allowing the parasite to complete its life cycle (Berdoy et al., 2000). Parasites can also indirectly alter host behaviour through the host compensatory response (Lefèvre et al.,

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2008) or through adaptive responses like sickness behaviours (Lopes et al., 2021; Tizard, 2008). For instance, Kirsten et al. (2018) showed that zebrafish, *Danio rerio*, reduce locomotion, social interactions and exploratory behaviours when the fish immune system is activated. Second, pre-existing individual behavioural differences may influence host susceptibility to infection (Koprivnikar et al., 2012; Santicchia et al., 2019; Wilson et al., 1993). For example, Wilson et al. (1993) found differences in the parasite fauna of pumpkinseed sunfish, *Lepomis gibbosus*, collected using minnow traps or seine nets, which select for different personality types. They suggested that behavioural differences, as assessed by susceptibility to each fishing method, predict the parasite fauna that a fish is likely to encounter, and thus, be infected with. Similarly, an experimental study showed that more exploratory and active tadpoles (*Lithobates sylvaticus*) are less susceptible to infection with trematodes (*Echinoparyphium* spp.), suggesting that personality differences are strong predictors of infection probability (Koprivnikar et al., 2012). However, distinguishing between these two hypotheses can be challenging in correlational studies (Barber & Dingemanse, 2010).

Experimental infections can be helpful to disentangle the causal direction between parasitism and behaviour. These protocols are often conducted in the laboratory by infecting a host with a known number and species of parasite(s) under controlled conditions (Hua et al., 2019; Sharp et al., 1992). While laboratory infections are informative, they do not capture the complexity of natural conditions, where co-infection (i.e. infection with more than one species of parasite) and environmental variations can alter the nature or the impact on hosts (Venter et al., 2022). For instance, mice (*Mus musculus*) co-infected with influenza-tuberculosis experience more severe disease than when infected with a single pathogen (Walaza et al., 2020). An alternative to experimental infection in laboratory settings involves infecting animals in seminatural environments (Mathieu-Bégné et al., 2022; Mauduit et al., 2022). Mauduit et al. (2022) successfully infected juvenile Chinook salmon, *Oncorhynchus tshawytscha*, with two myxozoan parasites by placing individuals in cages in a river. Despite the challenges presented by using seminatural infections (e.g. controlling individual parasite exposure), they are more representative of what occurs in the wild and allow individuals to experience fluctuating environmental conditions similar to what they experience during natural infection. In addition, measuring individual behaviour before and after an experimental infection can help us understand the degree to which initial behavioural traits modulate the impacts of infection on behaviour.

Body condition, an index of general health (Bolger & Connolly, 1989; Sánchez et al., 2018), is a state-dependent trait that influences and can be influenced by both individual behaviour and infection status. For instance, risk taking covaries with body condition in great tits, *Parus major* (Moiron et al., 2019). Body condition is often reduced in the presence of parasites (Lagrue & Poulin, 2015). Conversely, hosts in better condition may be able to tolerate more parasites and thus, a positive correlation between body condition and parasite load can also exist (Sánchez et al., 2018). Either way, body condition can provide insight into how parasitic infection affects host health, which can also be linked to behaviour. For instance, Thelamon (2023) found a negative relationship between parasite load and exploratory behaviours in fish hosts, but only for individuals in poorer body condition. Assessing how body condition changes during the course of an infection is thus an important metric to consider when exploring relationships between behaviour and parasitism.

Here, we explored the relationships between behaviour, parasite density and body condition in wild-caught pumpkinseed sunfish, *Lepomis gibbosus*. Pumpkinseed sunfish can be infected by numerous parasitic worms including trematodes, cestodes and nematodes in the wild (Chapman et al., 2015) and have been used in previous studies to measure aspects of trappability, boldness, exploration and cognition (Thambithurai et al., 2022; Thelamon, 2023; Wilson et al., 1993). In Laurentian lakes where this study was conducted, sunfish are commonly infected by at least two species of trematodes (e.g. *Uvulifer ambloplitis* and *Apophallus* sp.) causing black spot disease and one cestode, the bass tapeworm, *Proteocephalus ambloplitis*, causing liver disease. These parasites both have complex life cycles that require the sunfish to be depredated for their transmission (Fig. 1). We tested the same individuals' behaviour (i.e. exploration, activity, boldness) before and after an experimental infection in a lake with high parasite prevalence. We restricted fish movement in the lake by placing them in cages to allow natural co-infection. Specifically, our study had three main questions. (1) Are there consistent behavioural differences (i.e. personality) and behavioural syndromes in sunfish? (2) Does co-infection alter individual behaviour and covariance among behavioural traits? (3) How does co-infection affect body condition and behaviour? We were also able to test whether initial behavioural traits influence susceptibility to infection while in cages. When fish are uninfected, we expected to find consistent individual behaviour differences and a positive correlation between exploration, activity and boldness as these traits often form a behavioural syndrome (Herde & Eccard, 2013; Monceau et al., 2015). Following infection, we expected a reduced repeatability of behavioural traits because reducing interindividual variance in behaviour among fish that are infected should help parasites maximize their transmission to final hosts (Poulin, 2013). Following the same idea, behavioural syndromes should be strengthened by infection, and we expected an increase in mean exploration, activity and boldness at the population level because sunfish are intermediate hosts in the parasite life cycle, and increases in these traits and syndromes could result in more vulnerability to predation to facilitate parasite transmission (Berdoy et al., 2000). Body condition should be reduced in naturally infected fish, which could also influence correlations between infection and behaviour (Lagrue & Poulin, 2015; Thelamon, 2023).

METHODS

Animal Collection and Husbandry

Sampling and experiments were carried out at the Université de Montréal's Station de Biologie des Laurentides in Saint-Hippolyte, Québec, Canada (45°59′20″N, 74°0′W), between May and September 2022. We collected 72 pumpkinseed sunfish by seine net in Lake Cromwell (45°35′32″N, 73°35′44″W) on 30–31 May 2022. We kept subadult individuals measuring 4–5 cm (1–2 years of age, mean (±SE) mass: 2.22 ± 0.03 g, mean (±SE) standard length: 4.33 ± 0.02 cm) with few visible black spots (*Apophallus* sp. and *Uvulifer* sp.; minimum–maximum: 0–28; median: 2). Fish were transported back to the laboratory in buckets within 2 h of capture. Following capture, fish underwent a salt bath (3 g/litre) quarantine treatment to reduce fungal and bacterial infections. Fish were given two treatments of praziquantel (2 mg/litre) 3 days apart as a bath for 24 h to eliminate unencysted cestodes and trematodes (Bader et al., 2019). Fish were given a 2-week acclimatization period following these treatments before the behavioural experiments began.

Following the quarantine period, fish were housed in six tanks ($50.3 \times 24.1 \times 29$ cm, 45 litres), separated with a partition to ensure group stability ($N = 6$ per group, 12 groups in total) and reduce aggressive interactions. Groups were kept the same throughout all the experiments (before, during and after parasite

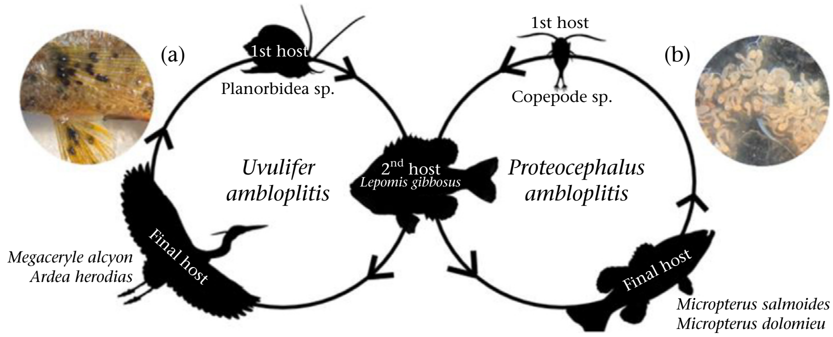

Figure 1. Life cycle of trematodes and cestodes infecting pumpkinseed sunfish in Laurentian lakes. (a) Trematodes (e.g. *Uvulifer ambloplitis* and *Apophallus* sp.) need multiple hosts for development and reproduction to occur: a freshwater snail (e.g. *Amnicola limosa*), a sunfish and a piscivorous bird (e.g. belted kingfisher, *Megaceryle alcyon*; great blue heron, *Ardea herodias*). (b) Bass tapeworms (*Proteocephalus ambloplitis*) are also endoparasites with a complex life cycle, which requires a planktonic crustacean like copepods, a small fish like the sunfish and a piscivorous fish, including smallmouth bass, *Micropterus dolomieu*, and largemouth bass, *Micropterus salmoides*, to complete its life cycle. In both cases, these parasites require the sunfish to be depredated to complete their transmission to their final hosts. Parasite photographs taken by A. Côté.

exposure). Water was supplied via a flow-through system pumped from nearby Lac Croche (45°59′24″N, 74°0′20″W), filtered and sterilized with ultraviolet light. Fish were fed 12 g of frozen bloodworms across all tanks every morning. Tanks were siphoned 1 h following feeding to remove uneaten food and excrement. PVC tubes and plastic plants were provided for environmental enrichment. Fish were maintained on a natural light cycle (14:10 h light:dark). We weighed and measured all fish (standard length) and quantified their parasite load via visual inspection of black spots on the left side of the fins and body. Black spot infection does not differ between the right and left side of a fish (De Bonville et al., 2024), so we quantified parasite intensity only on the left side to avoid double counting on unpaired fins and to minimize stress during inspection. To identify individuals, we implanted a unique colour code in the fish using a visible elastomer implant (VIE tags, Northwest Marine Technology, Anacortes, WA, U.S.A.). To minimize the impact of this procedure on fish behaviour, behavioural experiments started 7 days following tagging. Fish were feeding the next morning. Behavioural experiments were conducted before and after natural exposure to parasites in a lake.

Natural Parasite Exposure

Natural parasite exposure was carried out in Lake Cromwell where the prevalence of parasites is very high (78% black spot prevalence in similarly sized sunfish; Vigneault, 2024) using 10 galvanized steel cages (60 × 60 × 45 cm) with six fish per cage ($N = 60$). Since keeping fish in cages without risk of infection was not possible, two groups of six individuals ($N = 12$) were maintained in separate tanks in the laboratory to control for effects of time and habituation (Mauduit et al., 2022). These groups were also used to confirm the effectiveness of the antiparasite treatment. Shelters made of PVC tubes and plastic plants were attached to the bottom of the cages to reduce stress. Fish were fed with frozen bloodworms every day for the first week, and then every other day for the rest of the month at random times of the day to avoid anticipatory behaviour and encourage natural feeding on zooplankton, which is necessary for infection with cestode parasites. Control fish in the laboratory were also fed with frozen bloodworms at random times of the day. Cages were placed near the shore of the lake, about 1.5 m deep in the water in locations with enough water circulation to promote oxygenation and stable temperatures (see Appendix, Fig. A1). The cages were left for 30 days to allow natural exposure to parasites (7 July–5 August 2022) (Fig. 1). Caged fish were then brought back to the laboratory. Both caged fish and control fish were again treated in a salt bath for 24 h before being placed back in their holding tanks. Fish were weighed and measured again in the following days before behavioural trials (T3–T4), which started 7 days following their return to the laboratory.

Behavioural Experiments

All behavioural experiments were conducted between 0830 and 1300 hours to standardize hunger levels and other sources of diurnal variability. Four arenas were used to measure behavioural traits: exploration, activity and boldness. These traits were measured twice before (T1 and T2) and twice after (T3 and T4) parasite exposure in the lake. Two randomly selected fish from each of the 12 groups ($N = 10$ experimental groups; $N = 2$ control groups) were tested daily (24 fish per day). The time interval between trials (i.e. between T1 and T2 and between T3 and T4) was 7 days. Experimental tanks were filled with the same water as the holding tanks and changed between each test fish. Fish were transported from their holding tank to the test tank in a covered water-filled beaker. All experiments were filmed from above with a GoPro camera Hero 4 (GoPro, San Mateo, CA, U.S.A.) set at 24 frames/s.

Novel environment test

We used a novel environment test to record exploration and activity behaviour (Mazue et al., 2015). Before starting the trial, an individual was placed in a clear plastic arena (76 × 30 × 30 cm, water height 10 cm). Fish were given 1 min to acclimate to the environment before behaviour was recorded. We recorded exploration behaviour over 5 min as the percentage of surface area (%) covered by a fish in the new environment. One more minute was given where no behaviour was recorded. We then recorded fish activity as the distance (cm) moved in the tank over the next 5 min (Archard & Braithwaite, 2011). To make the tank a novel environment for each trial, we added small pieces of coloured tape to the walls of the experimental tank (T1: no tape; T2: red; T3: yellow and blue; T4: yellow and green). We calculated exploration from the *x* and *y* coordinates provided by Lolitrack 5 software (Loligo Systems, Tjele, Denmark) using R version 4.2.2 (R Core Team, 2022). We calculated the percentage of surface area covered (%) using the raw pixel position of each test fish in the videos. The resolution of videos used was 100 pixels. We transformed all pixels into 0 (absence of the fish) and 1 (presence of the fish) to calculate the surface area covered. Distance moved was calculated and extracted automatically using Lolitrack.

Shelter emergence test

We conducted a shelter emergence test 3 days later to assess behaviour along the bold–shy continuum (Brown et al., 2005). Individuals were transferred from the water-filled beaker into the middle of a 16 × 35 cm shelter area behind a closed sliding door for a 15 min habituation period. Two plastic plants were fixed to the bottom in the shelter area. When the habituation period was over, recording started, and the observer opened the sliding door by gently lifting the door. Emergence time (s) for the fish to leave the shelter area completely (i.e. whole body and caudal fin fully enter the open compartment) was recorded and extracted from the videos ($N = 288$) by one observer. Videos were examined out of order (i.e. not following trial order or testing order of individuals) to reduce observer bias.

Dissections

Following T4, we weighed and measured fish and counted visible black spots using previously described methods (Thambithurai et al., 2022). Fish were then euthanized with an overdose of clove oil and dissected within 2 min of being sacrificed. We inspected the liver, body cavity and digestive tract for moving (living) parasites. We made a distinction between parasites that were alive (moving) and those that were presumed dead (i.e. not moving) to account for any cestodes that were killed by the praziquantel treatment compared to those that were newly acquired during the natural parasite exposure. Brain and gills were examined for parasites as well. Afterwards, we crushed muscles to quantify encysted parasites (black spot, *Clinostomum marginatum*, or *Posthodiplostomum* sp.). Presence or absence of mature gonads was noted. All dissections were performed within a week after T4. To calculate parasite density, we used the number of black spots acquired during the natural parasite exposure period (i.e. the number counted post-infection minus the number counted pre-infection). For cestode density, we only considered the live adult form (i.e. plerocercoid) as this stage is infectious.

Statistical Analyses

All statistical analyses were performed in R version 4.2.2 (R Core Team, 2022). All variables were z-scaled (i.e. $(x - \bar{x})/\sigma$) before fitting the models to ease model interpretation and improve parameter estimation. We fitted our models within a Bayesian statistical framework using the packages ‘brms’ version 2.18.0 (Bürkner, 2017) and ‘rstan’ version 2.26.15 (Stan Development Team, 2021). We ran four Markov chain Monte Carlo (MCMC) chains each with 6000 iterations, sampling every iteration. We discarded the first 2000 iterations as a warm-up. We visually inspected chains to ensure that they were mixing well and that chains had converged (i.e. $\hat{R} < 1.02$) and ensured that we had an effective sample size >1000 for all parameters. Statistical significance of estimates from the models were inferred from whether credible intervals included zero. When we refer to the variable ‘treatment’ in our models, ‘control before’ ($N = 12$) refers to control fish from the two behavioural trials (T1 and T2), ‘uninfected’ ($N = 60$) refers to fish from T1 and T2 that would subsequently be naturally infected, ‘infected’ refers to the fish following parasite exposure and ‘control after’ refers to the control fish at T3 and T4.

Susceptibility to infection

To test whether behaviour influenced parasite infection within cages, we tested whether individual behaviour before parasite exposure (T1–T2) for each trait was related to black spot and cestode density acquired during parasite exposure. For each trait measured, we fitted one model with black spot density as the response variable and one model with cestode density as the response variable, behaviour before (T1–T2) as a fixed effect and fish identity and cage number as random factors (see Appendix, Table A1, model 1).

Personality and behavioural syndromes

To test for evidence of personality (i.e. traits were repeatable) and how co-infection affected repeatability, we built a multiresponse model with emergence time, surface area covered and distance moved as response variables. We visually inspected response variables and used q-q norm plots to assess assumptions of normality. Distance and emergence time were log-transformed. We included treatment (uninfected; infected; control) as a fixed effect for each response variable because we predicted that infection would change fish behaviour. We included fish identity ($N = 72$) and cage number ($N = 12$, 10 cages + 2 control groups) as random effects in the model (see Appendix, Table A1, model 2). Given that experimental infection could also impact within-individual variance (i.e. residual variance) differently across treatment groups, we fitted a heterogenous residual variance model. Our first model included tank as a fixed factor. We compared models using the ‘loo’ package version 2.5.1 (Vehtari et al., 2023) and decided to remove the variable tank (i.e. tanks used for experimental trials) as the model without tank was better fitted (loo difference: Δ-1.1 ± 1.5). We calculated repeatability for each treatment and overall from this model (see Appendix, equation A1, for repeatability calculation).

To test for the presence of behavioural syndromes (i.e. correlation among traits) and how co-infection affected these, we built a multiresponse model with the three behavioural traits measured as response variables. We included treatment (uninfected, infected) as a fixed effect for each response variable as we wanted to compare behavioural syndromes between the uninfected and infected state only. Therefore, we fitted a model without control fish. We included fish identity ($N = 60$) and cage number ($N = 10$) as random factors in the model (see Appendix, Table A1, model 3). To measure the average correlation for each trait, we pooled MCMC chains for the uninfected and infected states. To understand how changes resulting from co-infection covaried (i.e. correlations between random slopes of each trait) and how much each individual changed when infected (i.e. intercept–slope comparisons between traits), we fitted a multiresponse model with the three behavioural traits measured as response variables. We included treatment as both a fixed and a random effect. We estimated a random slope and intercept for treatment at the among-individual (fish identity) level (see Appendix, Table A1, model 4).

Body condition, parasite density and behaviour

To explore the relationships among body condition, parasite density and behaviour, we calculated body condition with the Fulton index (mass/length^3 (g/cm)) (Jakob et al., 1996). As mass and length were recorded following each behavioural trial, we calculated four measures of body condition for each fish. We used adjusted fish mass (i.e. fish mass minus parasite mass; Lagrue & Poulin, 2015) for the two measures (T3 and T4) of body condition following parasite exposure. For bass tapeworms, adult parasite average mass was 0.003 g and the larval form was 0.0008 g. We decided to exclude nematodes since the mass varied greatly between individuals and were rarely found in our fish samples ($N = 4$ fish). Black spot mass was considered too small to be subtracted from total mass (10^{-7} g). We calculated parasite density as the total number of parasites divided by the adjusted fish mass calculated before sacrifice. To test whether body condition changed during infection, we built a model with body condition as a response variable, cestode density, black spot density and treatment (uninfected; infected; control) as fixed factors and fish identity ($N = 72$) and cage number ($N = 12$) as random factors (see Appendix, Table A1, model 5).

To test whether co-infection and body condition influenced behaviour, we created two models on different subsets of data (uninfected versus infected). We included control fish for each model. For trials done when fish were uninfected (T1–T2), we built a multiresponse model with the three behavioural traits measured as response variables. We included body condition in the model as a fixed effect and fish identity ($N = 72$) and cage number ($N = 12$) as random factors (see Appendix, Table A1, model 6). For trials done when fish were infected (T3–T4), we built a multiresponse model with the three behavioural traits measured as response variables. We included cestode density, black spot density and body condition as fixed effects and fish identity ($N = 72$) and cage number ($N = 12$) as random factors (see Appendix, Table A1, model 7).

Ethical Note

Fish were collected and cared for with approval from the Université de Montréal's animal care committee and the Ministère de l'Environnement, de la Lutte Contre les Changements Climatiques, de la Faune et des Parcs (Comité de Déontologie de l'Expérimentation sur les Animaux; Permit number: 22-025, Collection permit number: 2022-05-16-1971-15-S-P). Fish were caught with a seine net ($N = 72$, 1–2 years of age) and rapidly removed from the net to minimize stress and injuries and brought back to the laboratory within 2 h of capture in aerated buckets. In housing tanks ($50.3 \times 24.1 \times 29$ cm, 45 litres), PVC tubes and plastic plants were provided as shelters and tanks were cleaned daily. Filtered water (temperature: 20 °C) from the lake was provided. We naturally exposed fish to parasite infection in the lake, with two parasites that commonly infect pumpkinseed sunfish at high prevalence at the study site. During experimental infection, we visited fish every day for a week, then every other day for the rest of the month to supplement their food supply. No fish died during behavioural experiments or during the caging experiment. All fish were ultimately humanely killed to look for internal parasites as part of this study. Fish were euthanized with a clove oil overdose mixed with lake water (4 ml/litre) (Fernandes et al., 2016). Individuals were left in the solution for 5 min, and heartbeat was inspected before dissection under a binocular microscope.

RESULTS

Susceptibility to Infection

There was a negative relationship between black spot density and distance moved (T1–T2) ($\beta = -0.43$, 95% CI $= -0.63, -0.24$) and between cestode density and distance moved (T1–T2) ($\beta = -0.30$, 95% CI $= -0.50, -0.10$): fish that initially moved shorter distances had a greater black spot and cestode density following parasite exposure. Similarly, emergence time at T1–T2 was negatively related to black spot density ($\beta = -0.27$, 95% CI $= -0.46, -0.08$) and cestode density ($\beta = -0.17$, 95% CI $= -0.36, 0.03$): fish that initially emerged faster had a greater black spot and cestode density following parasite exposure. We found no evidence that surface area covered at T1–T2 was correlated with black spot ($\beta = 0.13$, 95% CI $= -0.08, 0.34$) or cestode density ($\beta = 0.01$, 95% CI $= -0.20, 0.23$).

Personality and Behavioural Syndromes

Distance moved had moderate repeatability across trials regardless of infection status (Table 1). Emergence time had low repeatability in both states (Table 1). Percentage of surface area covered had moderate repeatability in the uninfected state but low repeatability in the infected state (Table 1). Repeatability was not significantly different between uninfected and infected fish for emergence time, surface area covered or distance moved (see Appendix, Fig. A2). Given the small number of control fish tested ($N = 12$), the repeatability was difficult to reliably estimate for control fish. Within- and among-individual variance also decreased following parasite exposure for distance moved and surface area covered (see Appendix, Fig. A3).

We found strong evidence of a correlation between distance moved and surface area covered (correlation = 0.63, 95% CI = 0.23, 0.90): uninfected fish that moved a greater distance also covered more surface area. This correlation declined when fish were infected, resulting in an overlap of the 95% CI with 0 (correlation = 0.10, 95% CI = −0.48, 0.64). We found no correlation between distance moved and emergence time when fish were uninfected (correlation = −0.47, 95% CI = −0.86, 0.13) or infected (correlation = −0.40, 95% CI = −0.83, 0.21). Similarly, we found no correlation between emergence time and surface area covered when fish were uninfected (correlation = −0.36, 95% CI = −0.82, 0.24) or infected (correlation = −0.38, 95% CI = −0.84, 0.30). Correlations did not differ between uninfected and infected states for these among-individual correlations (see Appendix, Fig. A4).

When looking at how changes in behavioural traits covaried after co-infection, we found a few nonsignificant correlations. Individuals who increased their surface area coverage across trials tended to decrease their emergence time following co-infection (Fig. 2a). Fish that decreased their emergence time also tended to increase the distance they moved (Fig. 2b). Fish that increased their surface area coverage also tended to increase the distance they moved (Fig. 2c). Although the correlation between fish personality (i.e. intercept) and changes in fish personality across treatments (i.e. slope) were not significantly related, we did find that fish with longer emergence times tended to decrease their emergence time following co-infection (Fig. 2d). Similarly, fish that covered less surface area showed an increase in their surface area coverage after natural parasite exposure (Fig. 2e). Fish that moved shorter distances also tended to increase the distance they moved following co-infection (Fig. 2e).

Differences in Average Behavioural Responses

Control fish (T1–T2) and uninfected fish (T1–T2) did not show different behavioural responses before natural exposure to parasites (Fig. 3). For both control and infected fish, when comparing behaviour before (T1–T2) and after (T3–T4) exposure to parasites, mean emergence time decreased and mean surface area covered increased (Fig. 3). In contrast, mean distance moved did not change over time for control and infected fish (Fig. 3). Surface area covered and emergence time differed between control and infected fish, but distance moved did not (Fig. 3).

Relationships Between Co-infection, Body Condition and Personality

Change in body condition did not differ significantly between uninfected and infected fish ($\beta = 0.27$, 95% CI $= -0.06, 0.61$) or between control fish at T1–T2 and at T3–T4 ($\beta = 0.41$, 95% CI $= -0.13, 0.95$). However, cestode density and body condition were negatively correlated ($\beta = -0.18$, 95% CI $= -0.33, -0.04$), but black spot density and body condition were not ($\beta = -0.13$, 95% CI $= -0.29, 0.02$). Body condition was positively correlated with distance moved when fish were uninfected ($\beta = 0.22$, 95% CI $= 0.04, 0.40$) or infected ($\beta = 0.25$, 95% CI $= 0.09, 0.40$), and distance moved decreased with black spot density (Fig. 4). Body

Table 1
Repeatability (*R*) estimates and 95% credible intervals (CIs) for emergence time (s), surface area covered (%) and distance moved (cm) for pumpkinseed sunfish overall (across trials 1–4 regardless of infection, *N* = 72), for uninfected sunfish (trials 1–2, *N* = 60) and for naturally infected sunfish (trials 3–4, *N* = 60)

Trait	Overall		Uninfected		Infected	
	R	95% CI	*R*	95% CI	*R*	95% CI
Log emergence time (s)	0.15	0.002, 0.35	0.15	0.002, 0.37	0.15	0.002, 0.38
Surface covered (%)	0.24	0.01, 0.45	0.29	0.07, 0.49	0.18	0.003, 0.42
Log distance moved (cm)	0.31	0.08, 0.50	0.33	0.12, 0.52	0.29	0.04, 0.51

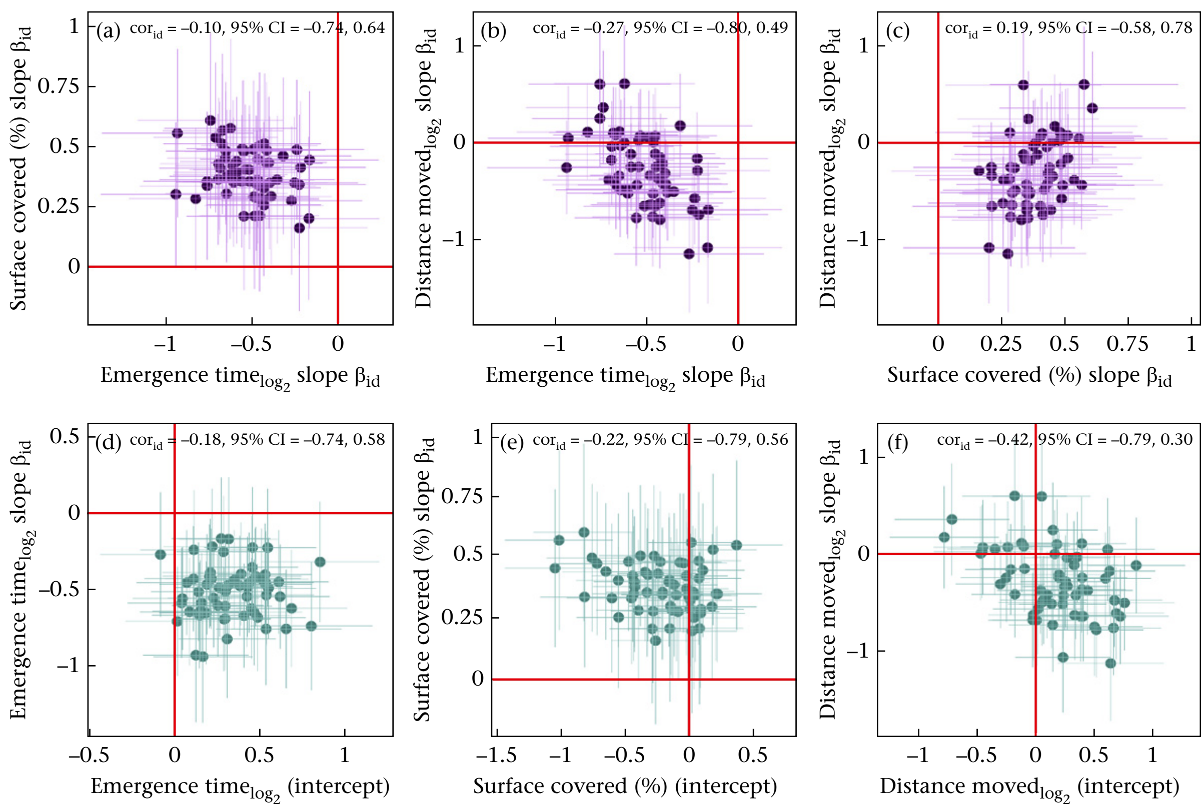

Figure 2. Correlation (cor$_{id}$) of slopes (β$_{id}$) of changes at the individual level (fish ID) ± SE between (a) emergence time (s) and surface area covered (%), (b) emergence time (s) and distance moved (cm) and (c) surface area covered (%) and distance moved (cm) of pumpkinseed sunfish following natural parasite exposure. Purple dots are individuals (*N* = 60); purple lines are standard errors; red lines denote average behaviour. For both axes, points above the red line (>0) denote a positive change in the slope and points below (<0) denote a negative change. All variables are *z*-scaled. Correlations (cor$_{id}$) between (d) changes in emergence time and average emergence time, (e) changes in surface area covered and average surface area covered and (f) changes in distance moved and average distance moved, with their standard errors, of pumpkinseed sunfish following natural parasite exposure. Green dots are individuals (*N* = 60); green lines are standard errors. On the *X* axis, points above the red line (>0) denote above-average behaviour and points below (<0) denote below-average behaviour. On the *Y* axis, points above the red line denote a positive change in the slope of a trait and points below denote a negative change. All variables are *z*-scaled.

condition was not related to surface area covered or emergence time when fish were uninfected (surface area covered: β = 0.11, 95% CI = −0.07, 0.29; emergence time: β = −0.01, 95% CI = −0.18, 0.15) or infected (surface area covered: β = −0.07, 95% CI = −0.22, 0.08; emergence time: β = 0.12, 95% CI = 0.06, 0.29).

DISCUSSION

We successfully infected wild pumpkinseed sunfish with trematodes causing black spot disease and bass tapeworms causing liver disease using a seminatural experimental infection in a lake with high parasite prevalence. Most fish were infected with both parasites (*N* = 46 co-infected fish out of 60; see Appendix, Fig. A1). Surprisingly, our cage design did not control for how initial behaviour influenced susceptibility to infection. Indeed, some individuals were less or more likely to be infected given their initial behavioural traits. Following infection, parasite density was also linked to behavioural traits. Thus, we suggest that behaviour and parasitism are tightly linked, and both mechanisms (behaviour influencing parasitism and parasitism influencing behaviour) could simultaneously explain relationships between infection and behaviour in wild animals.

Susceptibility to Infection

Despite our initial assumption, we found that initial behavioural traits were linked to susceptibility to parasite infection within our cages. Fish that were less active and/or bolder in our initial behavioural trials (at T1–T2) were more likely to be infected with both parasites (black spots and cestodes) following lake exposure. This suggests that infection risk was not the same among individuals within a cage. Bolder fish had higher black spot density

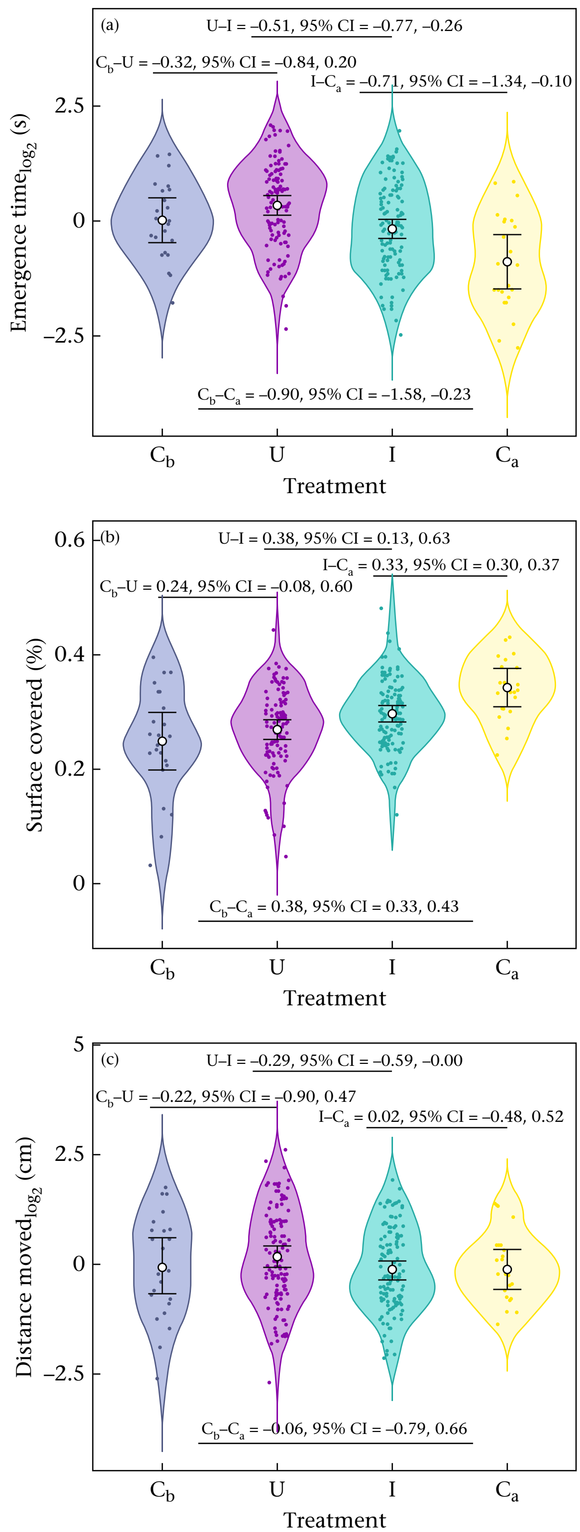

Figure 3. Contrasts in mean (±95% credible interval) responses of control (C_b, $N = 12$) and uninfected (U, $N = 60$) pumpkinseed sunfish in trials 1–2 and of control (C_a, $N = 12$) and infected (I, $N = 60$) pumpkinseed sunfish in trials 3–4 for (a) emergence time (s), (b) surface area covered (%) and (c) distance moved (cm). Coloured dots are observed data; black lines are credible intervals predicted by the model; white dot is the mean behaviour predicted by the model. Distance moved and emergence time are log-transformed.

than shyer fish, a result which is consistent with a study showing that bolder pumpkinseed sunfish had greater accumulation of black spot in natural settings (Wilson et al., 1993). Despite being confined to a cage, bold fish may nevertheless inspect more of the cage more frequently, exposing them to a higher risk of infection (Sih & Del Giudice, 2012). Boldness was also linked to cestode density. Bolder fish are known to forage more to support faster growth rates compared to shyer individuals (Barber & Dingemanse, 2010; Wilson & Krause, 2012). In our case, this may imply increase feeding on infected copepods and thus, greater cestode infections. Cestodes can also alter host energy metabolism, which may lead to increased foraging in some systems (Cunningham et al., 1994). Less active fish were also more susceptible to black spot and cestode infection. Individuals with high activity levels may have been in better condition (Koolhaas, 2008) and thus, more resistant to parasite infection. This is consistent with the positive relationship between body condition and activity we found. Behaviour and immunity can be tightly linked, and the correlations we found might be representative of an individual's immune system (Capitanio, 2008; Friedman, 2008), which could influence infection risk. For instance, an individual's infection history could also influence infection risk within the cages: we found a moderate positive correlation between black spot acquired before and after the parasite exposure (Appendix, Fig. A5), suggesting that some individuals may be more resistant than others to this infection. Similarly, we found a moderate positive correlation between dead cestodes (previously killed with praziquantel) and live cestodes following infection (Appendix, Fig. A6). These differences in terms of the immune system should be explored in future studies to better understand the relationship between behaviour and infection risk in pumpkinseed sunfish. Sex of individuals and social environment could also have played a role in these initial behavioural and immune system differences (Campbell et al., 2021). For instance, infected male guppies, *Poecilia reticulata*, are able to sustain high activity levels when shoaling with females to 'hide' their sickness (Jog et al., 2022), and female guppies can increase their boldness in the presence of males (Piyapong et al., 2010). Nevertheless, our results suggest that some behavioural traits can increase infection risk, even in small, confined places like our caging design. These findings should be considered carefully when performing natural parasite exposure in cages to study behavioural variation.

Personality and Behavioural Syndromes in Sunfish

As expected, in a context without infection (i.e. before the cage experiment), we found that sunfish showed boldness, exploratory behaviours and activity levels that were repeatable with standardized tests. These aspects of animal behaviour are generally repeatable across contexts: a meta-analysis found that the average repeatability for a wide variety of behaviours and taxa is around 0.37 (Bell et al., 2009). In our case, we found moderate repeatability for exploration and activity ($R = 0.29–0.33$) and low repeatability for boldness ($R = 0.15$). This latter result is consistent with another study on pumpkinseed sunfish where boldness measures were not repeatable using a different emergence shelter test (Thelamon, 2023). When there is little variation in a trait within a population, repeatability estimates will necessarily be low. Indeed, as a species, sunfish are known for their boldness: they are easily caught with hooks and in minnow traps, readily approach divers and snorkellers and thrive in captivity (Thambithurai et al., 2022; Wilson et al., 1993). As such, we suggest that laboratory shelter tests such as ours might not be useful measures of boldness in this species because there is not enough natural interindividual variation in this behaviour. Indirect measures of boldness like trappability may more accurately reflect differences in boldness in pumpkinseed sunfish and should be explored in future studies (Thambithurai et al., 2022; Wilson et al., 1993).

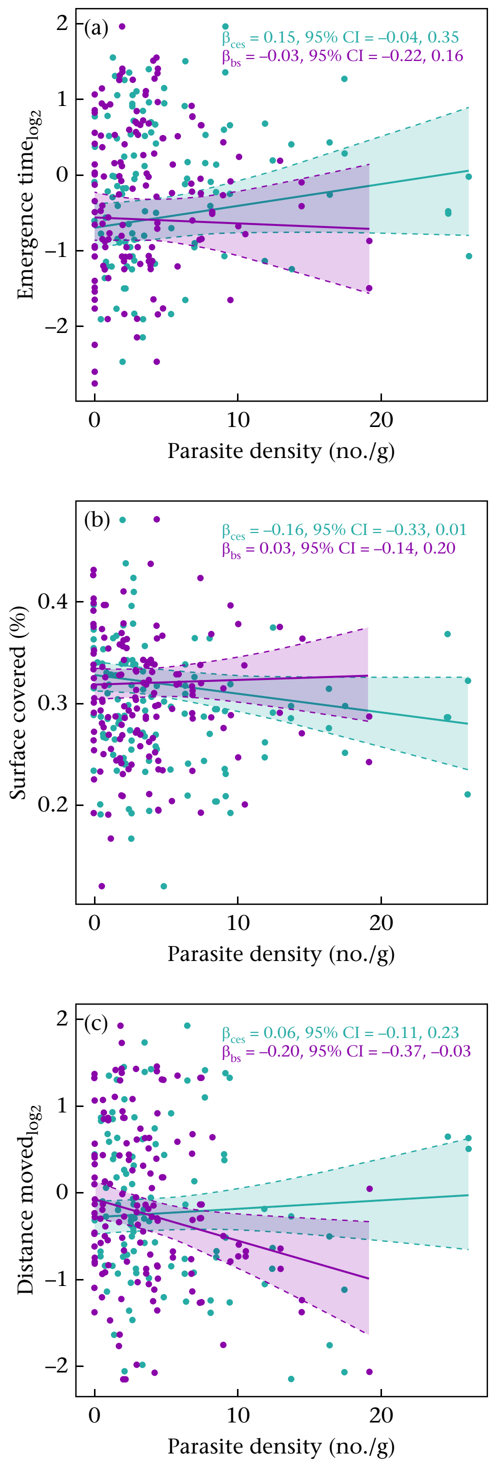

Figure 4. Effect of cestode (ces, green lines) and black spot (bs, purple lines) density on (a) emergence time (s), (b) surface area covered (%) and (c) distance moved (cm) of pumpkinseed sunfish after being naturally exposed to parasites. Solid line is the slope and dotted lines are the credible intervals predicted by the model; solid dots are the raw data, with two dots for each individual ($N = 72$). Distance moved and emergence time are log-transformed.

When looking at changes in behavioural traits through time, we found that average measures of exploration and boldness increased in T3 and T4 compared to T1 and T2 for both infected and control fish, suggesting that habituation may have played a more important role in population level changes in these behaviours than parasite infection. Habituation to novelty can occur when measuring the same behaviour multiple times on the same individual, especially in laboratory settings (Blumstein, 2016). We tried to prevent this from happening by slightly modifying the environment at every trial (i.e. changing wall colours with coloured tape). Nevertheless, our fish appear to have rapidly habituated to repeated tests. This shows how quickly habituation can occur in wild animals after spending extensive time in the laboratory and the difficulty of measuring traits such as exploration with repeated measures in laboratory settings. Despite this, measures of activity did not change across trials for control fish, suggesting individuals did not habituate to this test. Thus, changes in activity levels following caging likely represent the experimental treatment rather than an effect of habituation over time.

We found evidence of behavioural syndromes between traits when sunfish were uninfected, as we expected. We found that more active fish covered more surface area (i.e. were more exploratory). Correlations between activity–boldness and exploration–boldness were low, despite the fact that these traits often form a behavioural syndrome (Kudo et al., 2021; Mazue et al., 2015). As noted above, we found little variation between individuals for boldness, which could explain the low covariance with the other traits. Activity and exploration were positively correlated, which is also documented in many systems (Herde & Eccard, 2013; Monceau et al., 2015). Such a syndrome in pumpkinseed sunfish could suggest a trade-off between foraging and predation in a context without infection (Smith & Blumstein, 2008). More active and exploratory fish could increase their fitness by increasing foraging rates, but this could lead to increasing predation risk. Less exploratory and less active individuals, in contrast, could reduce foraging rates but have a higher chance of survival. However, this syndrome might no longer be adaptative if environmental conditions and/or individual internal states change.

Personality and Behavioural Syndromes in the Context of Infection

Few studies have investigated whether parasitism influences repeatability and behavioural syndromes. Following parasite exposure, we found that both repeatability and the strength of relationships between behavioural traits (i.e. behavioural syndromes) seem to be reduced. Exploration and activity became less repeatable (exploration: $R = 0.29–0.18$; activity: $R = 0.33–0.29$), and the behavioural syndrome between activity and exploration was no longer significant. Parasites may reduce variance among individuals to maximize their transmission. Indeed, if all infected individuals in a population act more similarly, and thus predictably, they may be more likely to be caught (Poulin, 2013). For example, Coats et al. (2010) found reduced repeatability in activity for amphipods (*Paracalliope fluviatilis*) infected with a trematode (*Microphallus* sp.). Our findings support these results. We found that for both exploration and activity, within- and among-individual variances decreased following infection (see Appendix, Fig. A3). We suggest that behavioural syndromes might be context dependent, with individuals exhibiting more plasticity (i.e. less consistency) depending on state-dependent or environmental factors such as infection (Kudo et al., 2021). Repeatability and correlations among traits were not significantly different before and after parasite exposure. Indeed, our sample size was small and repeated measures were limited to four trials, which may limit our statistical power. Nevertheless, these results suggest that co-infection could potentially affect the repeatability of behaviour by reducing both within- and among-individual variance and alter correlations between traits by decreasing consistency in behaviour.

Parasite Infection Increases Behavioural Plasticity

When looking at how initial behaviour affected how much individuals changed after being naturally exposed to parasites, we found a negative correlation between activity level and the change in activity. In simpler terms, more active fish tended to have the strongest reduction in activity, while less active fish tended to have the strongest increase in activity after infection. This suggests that infection did not change average activity but did affect how individuals plastically responded to parasite infection. This result may be explained by parasites reducing the variability among individuals, such that individuals converge on a common mean activity level when parasitized to facilitate parasite transmission (Poulin, 2013), which is consistent with the reduced variance among and within individuals that we found. Indeed, very active individuals may be able to escape predation by some predators (Smith & Blumstein, 2010), while freezing may limit predation risk by avian predators in some species (Berger & Aubin-Horth, 2020). Although the correlation was moderate in magnitude, it was not statistically significant. However, estimating such effects is notoriously challenging given the low statistical power in many studies. Despite this limitation, this pattern could provide clues on the mechanism behind how co-infection impacts behaviour in infected sunfish and can create behavioural plasticity in this context.

Cestode Infection Reduces Host Body Condition

We found that fish body condition did not differ before and after parasite exposure, suggesting that the average body condition for all fish did not vary through time. However, body condition was negatively correlated with cestode density. This suggests that cestode infection is energetically costly to hosts and that costs increase when more parasites are present. Infection can impose energetic costs on hosts, and cestodes can have a debilitation effect on pumpkinseed sunfish through tissue and liver damage (Guitard et al., 2022). Heavy infections of cestodes have been linked to decreased metabolic rates and escape responsiveness in naturally infected sunfish, which could lead to increased vulnerability to predators (Guitard et al., 2022). Cestode infection was also linked with decreased body condition in three-spined stickleback, *Gasterosteus aculeatus* (Bagamian et al., 2004), which is consistent with what we found. Black spot infection also had a small negative effect on body condition, yet this was not statistically significant. Lemly and Esch (1984) found that body condition of bluegill sunfish, *Lepomis macrochirus*, was reduced with an accumulation of more than 50 encysted black spots, which is more than the mean and median number of trematodes gained by our fish (mean: 8.4; median: 6.5). Leaving the cages in the lake for a longer period to allow higher infection levels may provide more clues as to the impacts of both types of infection on host body condition.

Reduced Activity Levels with Increased Black Spot Density

We found that activity was positively correlated with body condition in both contexts (uninfected versus infected), but negatively correlated with black spot density in naturally infected fish. This suggests that fish that accumulated more trematodes had a greater reduction in activity levels, which could also be associated with reduced body condition. However, we did not see any significant interaction between body condition and black spot density (Appendix, Table A2). Nevertheless, black spots could potentially reduce body condition as explained above. Activity levels are often positively correlated with general health (Cavigelli, 2005) and metabolic rates (Careau et al., 2008), explaining why body condition was also positively correlated with activity in our study. We suggest that infected fish decrease their activity levels with increased black spot density, possibly as an energy-saving strategy when fighting infection (Grossberg et al., 2011). Sickness behaviours, including lethargy, anorexia and anxiety, occur when there is an immune system activation to fight infection (Lopes et al., 2021; Tizard, 2008). For instance, immune stimulation causes reduced locomotion, social interactions and exploratory behaviours in zebrafish (Kirsten et al., 2018). Indeed, Lemly and Esch (1984) found that trematode infection resulted in increased resting metabolic rates in sunfish during the acute phase of the infection, suggesting that trematode encystment is energetically costly to hosts. Although this result does not follow our initial prediction, this decrease in activity could be important for fighting infection while also increasing vulnerability to avian predators. For instance, in guppies, less active individuals are more at risk of predation by piscivorous birds (Smith & Blumstein, 2010). Whether less active individuals are more susceptible to predation in this species remains to be tested.

Conclusions

We have shown that co-infection impacts pumpkinseed sunfish behaviour and overall health and that behaviour can enhance susceptibility to infection. More specifically, we found that less active and bolder individuals were more likely to be infected with both parasites. We also found that cestode infection decreased fish body condition, suggesting an energetic cost of this infection on hosts. However, parasite infection also seemed to play a role in creating behavioural variation. We found that black spot infection decreased activity levels, possibly because of immune activation or tissue damage, which has implications on the vulnerability of infected fish to avian predators. Note, however, that environmental conditions other than parasite exposure likely differed between individuals kept in cages and the control fish that remained in the laboratory (i.e. fluctuations in temperature, dissolved oxygen, food availability). Studies should be designed explicitly to test both hypotheses simultaneously: experimental infections in the laboratory and in natural settings to look at susceptibility to infection. Overall, our results contribute valuable insights into the complexity of health and behavioural changes caused by co-infection, emphasizing the need to investigate animal behaviour and infection more in experimental and natural contexts.

Author Contributions

Maryane Gradito: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing e review & editing. **Frédérique Dubois:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Daniel W. A. Noble:** Formal analysis, Visualization, Writing – review & editing. **Sandra Ann Binning:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Data Availability

All data presented in this study are available on the Figshare data repository (https://doi.org/10.6084/m9.figshare.24185637.v1).

Declaration of Interest

None.

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Appendix

Boldness is measured as emergence time (s), activity is measured as distance moved (cm) and exploration is measured as surface area covered (%). Sigma is the residual variance from the model. Model 1 ($N = 120$ observations) examined the influence of behaviour on infection risk (one model for black spot density and one for cestode density for each trait). Model 2 ($N = 288$ observations) was used to find evidence of personality (i.e. repeatability) and mean contrast between uninfected ($N = 60$, T1–T2), infected ($N = 60$, T3–T4) and control fish ($N = 12$, T3–T4) to determine whether habituation influenced behavioural changes. This was a heterogenous residual variance model. Model 3 ($N = 264$ observations) was used to estimate behavioural syndromes for uninfected ($N = 60$, T1–T2) and infected fish ($N = 60$, T3–T4). This was also a heterogenous residual variance model. We excluded control fish as we were not interested in this group and because of the group size, the model lost statistical power. Model 4 ($N = 264$ observations) tested for how changes resulting from co-infection covaried (i.e. correlations between random slopes of each trait) and how much individuals changed when infected (i.e. intercept–slope comparisons between traits). Treatment (uninfected, infected) was included as both a fixed and a random effect. Model 5 ($N = 288$ observations) examined the effect of cestode density, black spot density and treatment (uninfected, infected, control) on body condition. We wanted to determine whether body condition changed during co-infection. Model 6 ($N = 144$ observations) examined the effect of body condition on behaviour for uninfected individuals ($N = 72$, T1–T2). Model 7 ($N = 144$ observations)

Table A1
Detailed description and formulas of our multiresponse models using a Bayesian framework

Models	Formula
Model 1	Black spot/cestode density ~ behaviours before + (1 | ID_fish) + (1 | cage)
Model 2	Boldness/activity/exploration ~ 1 + treatment + (−1 + treatment |q| ID_fish) + (1|cage), sigma ~ −1 + treatment (uninfected; infected; control)
Model 3	Boldness/activity/exploration ~ 1 + treatment + (−1 + treatment |q| ID_fish) + (1|cage), sigma ~ −1 + treatment (uninfected;infected)
Model 4	Boldness/activity/exploration ~ 1 + treatment + (treatment |q| ID_fish) + (1 | cage), sigma ~ −1 + treatment (uninfected; infected)
Model 5	Body condition ~ 1 + cestode density + black spot density + treatment + (1 | ID_fish) + (1 | cage) (uninfected; infected; control)
Model 6	Boldness/activity/exploration ~ 1 + body condition + (1|ID_fish) + (1|cage)
Model 7	Boldness/activity/exploration ~ 1 + cestode density + black spot density + body condition + (1|ID_fish) + (1|cage)

examined the effect of cestode density, black spot density and body condition on behaviour of infected fish. Control fish were included in model 6 and model 7.

Repeatability

Repeatability (R) for each trait was calculated as follows:

$$R = \frac{\sigma^2_{\mathrm{ID}t}}{\sigma^2_{\mathrm{ID}t} + \sigma^2_{\mathrm{cage}} + \exp\,(\sigma_R)^2{}_t} \quad \text{(A1)}$$

From equation (A1), $\sigma^2_{\mathrm{ID}t}$ is the variance among fish for t, σ^2_{cage} the variance among cages and $\exp\,(\sigma_R)^2$ is the log-transformed standard deviation squared for t (i.e. residual variance), where t is the treatment state (uninfected; infected). Given that we estimated separate residual variances for each treatment, we were able to calculate repeatability across treatments. We also calculated an overall (i.e. marginal repeatability) by pooling the posterior distributions of each $\exp\,(\sigma_R)^2{}_t$ and $\sigma^2_{\mathrm{ID}t}$.

We found no living parasites in the abdominal cavity or body tissues of control fish (i.e. those that stayed in the laboratory), indicating that the praziquantel treatment was effective at killing unencysted helminths within the first week of capture. Fish were successfully infected with parasites following 1 month in the lake cages. The most abundant species found were trematodes causing black spot disease (*Apophallus* sp. and *Uvulifer* sp.; range 0–36; median = 6.5) and bass tapeworms (*Proteocephalus ambloplites*; range 0–36; median = 4) causing a liver disease. Across all cages, infected fish gained, on average, 8.4 black spots and 3.4 cestodes over the course of the infection (Fig. A1). No *Posthodiplostomum* sp. was found. Only one fish was infected with *Clinostomum marginatum*, but this parasite was not included in the subsequent analyses because of its low number. Not all fish were co-infected with both parasites: two fish had cestode infection only but no black spot, while 12 fish did not have any moving cestodes but were infected with black spots.

We suspected an interaction between body condition and densities of both parasites, so we fitted models to test this. We wanted to determine whether there was an interaction between the two types of parasites, and we built a new model with that interaction. Neither interaction was significant, so we excluded them in our final model.

Table A2
Multiresponse models looking at interactions between variables for each trait

Response variable	Interaction	Estimate	95% CI
Boldness	Black spot∗cestode	0.01 ± 0.11	[−0.20,0.24]
Exploration	Black spot∗body condition	0.17 ± 0.10	[−0.03,0.37]
Activity	Cestode∗body condition	0.02 ± 0.11	[−0.20, 0.24]
Boldness	Black spot∗cestode	0.06 ± 0.11	[−0.15,0.28]
Exploration	Black spot∗body condition	−0.03 ± 0.09	[−0.21,0.16]
Activity	Cestode∗body condition	−0.11 ± 0.10	[−0.31,0.08]
Boldness	Black spot∗cestode	0.11 ± 0.14	[−0.16,0.38]
Exploration	Black spot∗body condition	0.10 ± 0.12	[−0.34,0.13]
Activity	Cestode∗body condition	0.07 ± 0.13	[−0.18,0.32]

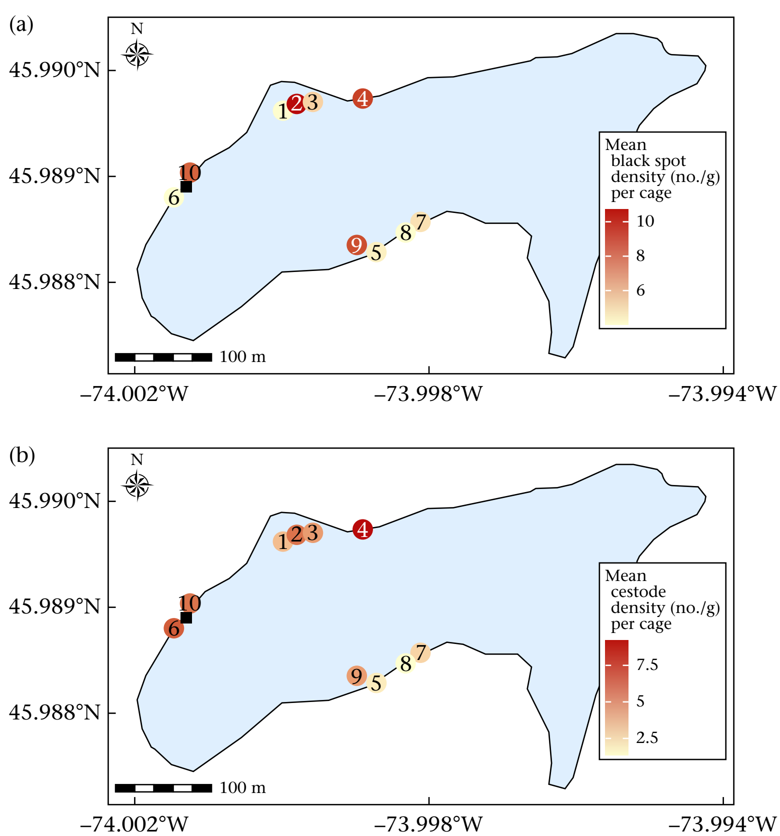

Figure A1. Mean (a) black spot and (b) bass tapeworm (b) density per cage ($N = 10$) placed at different sites around Lake Cromwell, Station de Biologie des Laurentides, Québec, Canada. Map coordinates are expressed in decimal degrees. Black square is the water entry point (boat dock).

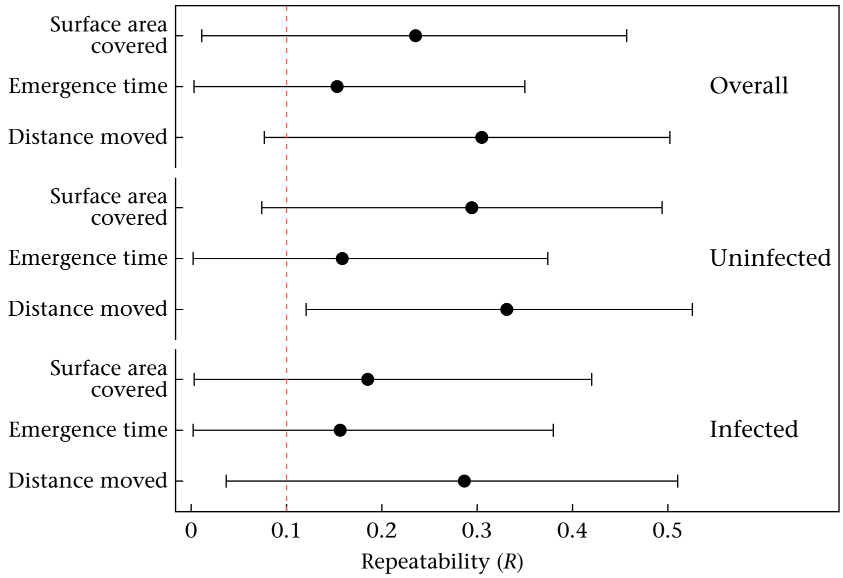

Figure A2. Repeatability estimates and their 95% credible intervals for the three behavioural traits (emergence time (s), surface area covered (%), distance moved (cm)) of pumpkinseed sunfish measured overall and within each treatment (overall, $N = 72$; uninfected, $N = 60$; infected, $N = 60$). Overall: behaviour of all fish regardless of treatment (trials 1–4); Uninfected: behaviour of individuals before natural exposure to parasites (trials 1–2); Infected: behaviour of individuals after natural exposure to parasites (trials 3–4). Black dots are repeatability estimates and black lines are the credible intervals from the model. Red dotted line denotes low repeatability ($R = 0.10$).

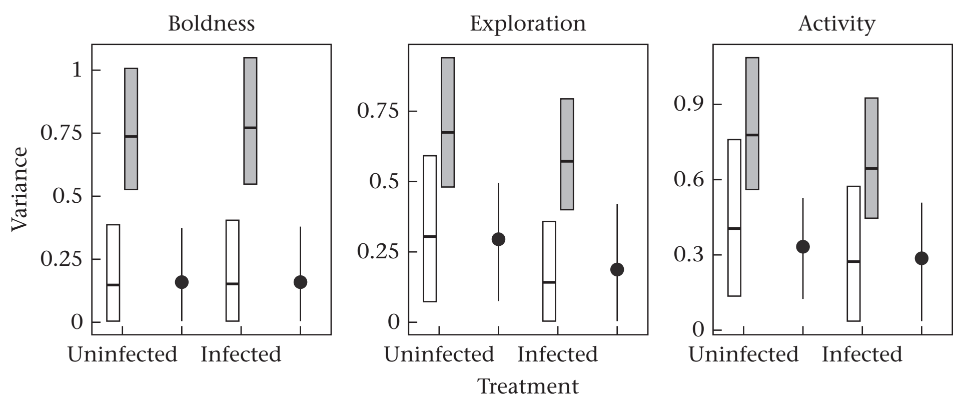

Figure A3. Within-individual variance (grey bars) and among-individual variance (white bars) of pumpkinseed sunfish before (uninfected) and after (infected) natural parasite exposure. Repeatability was calculated from the posterior distribution of model 2 and is represented by the black dot (mean) and black lines (95% credible interval, CI) next to the bars. The black line within each bar is the mean variance and the range of the bar is the 95% CI. Variance from cages was not included as it was constant in the repeatability equation (A1).

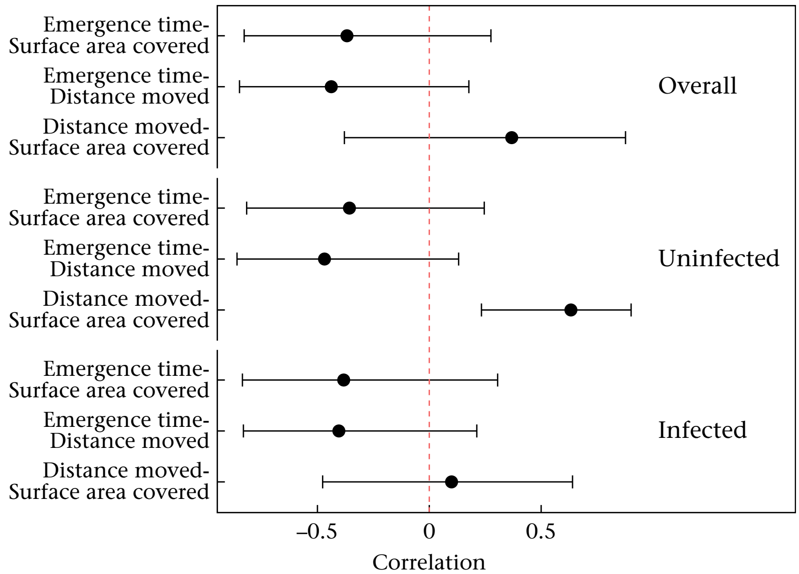

Figure A4. Correlation coefficients and their 95% credible intervals between the three behavioural traits (emergence time (s), surface area covered (%), distance moved (cm)) of pumpkinseed sunfish measured overall and within each treatment (overall, $N = 72$; uninfected, $N = 60$; infected, $N = 60$). Overall: repeatability regardless of infection status (trials 1–4); Uninfected: before natural parasite exposure (trials 1–2); Infected: after natural parasite exposure (trials 3–4). Black dots are correlation estimates and black lines are the credible intervals from the model. Red dotted line is zero in the distribution (if the credible intervals cross it, the correlation is not significant).

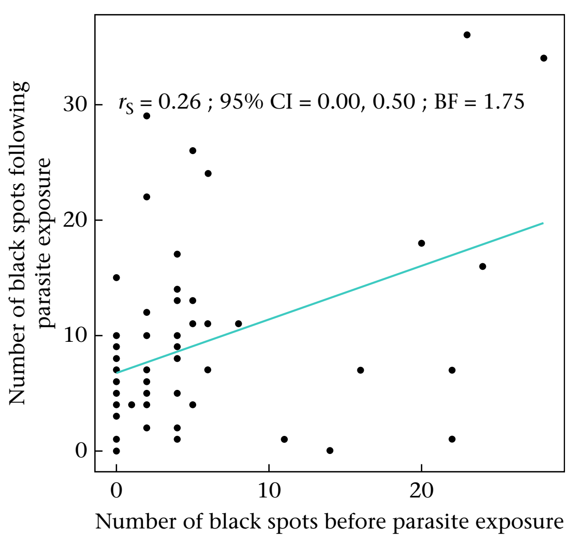

Figure A5. Spearman correlation performed in a Bayesian framework between the number of black spots acquired before and following parasite exposure. Black dots are individuals from trial 4 (following dissection) ($N = 60$). Green line is the Spearman correlation (r_S) between both variables. BF (Bayes factor) of 1.75 suggests moderate evidence of a positive correlation between both variables. The function 'cor_test' from the package 'correlation' was used in R version 4.2.2.

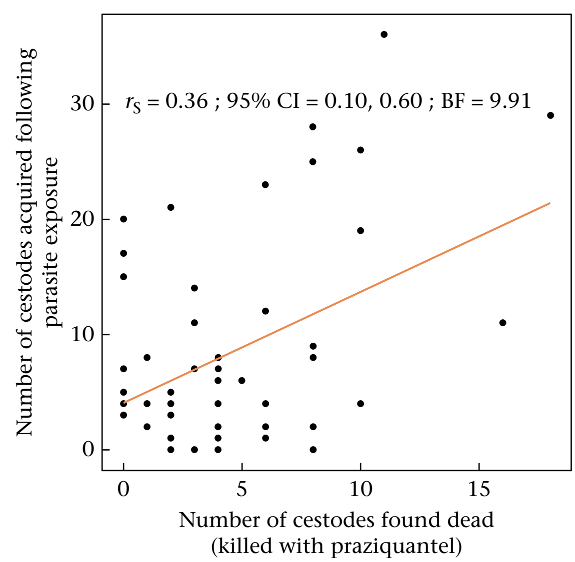

Figure A6. Spearman correlation performed in a Bayesian framework between the number of cestodes found dead and acquired following parasite exposure. Black dots are individuals from trial 4 (following dissection) ($N = 60$). Orange line is the Spearman correlation (r_S) between both variables. BF (Bayes factor) of 9.91 suggests strong evidence of a positive correlation between both variables. The function 'cor_test' from the package 'correlation' was used in R version 4.2.2.